MGM for I2I

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Overview

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Initial idea of MGM

Training

- Generate $X_0 = G_\theta(Z)$ and sample real X_1 independently.
- A vector field predicts displacement from a symmetric observation of the two endpoints.
- Update the generator using this learned distributional discrepancy.

Inference

$$Z \sim \mathcal{N}(0, I), \qquad X = G_\theta(Z).$$

No integration of the field is required.

The symmetry at the midpoint

If the independent endpoint distributions agree,

$$\mathbb{E}[X_1 - X_0 \mid (X_0 + X_1)/2] = 0.$$

Midpoint loss

$$X_{1/2} = \frac{X_0 + X_1}{2}.$$

$$D_{\mathrm{mid}}(p_\theta, p_{\mathrm{data}}) = \mathbb{E}\left\|\mathbb{E}[X_1 - X_0 \mid X_{1/2}]\right\|^2.$$

Learn the unknown conditional mean with a vector field:

$$\begin{aligned}&\mathbb{E}[2\langle f(X_{1/2}), X_1 - X_0\rangle - \|f(X_{1/2})\|^2] \\ &\quad = D_{\mathrm{mid}} - \mathbb{E}\left\|f(X_{1/2}) - \mathbb{E}[X_1 - X_0 \mid X_{1/2}]\right\|^2.\end{aligned}$$

$$\boxed{\min_\theta D_{\mathrm{mid}} = \min_\theta \max_f \mathbb{E}[2\langle f(X_{1/2}), X_1 - X_0\rangle - \|f(X_{1/2})\|^2].}$$

The maximum occurs at $f^*(x) = \mathbb{E}[X_1 - X_0 \mid X_{1/2} = x]$.

MGM, Sec. 3.1 and Proposition 3.3. The variational identity uses conditional expectation and completing the square.

Flipped interpolation and loss

$X_0 = G_\theta(Z)$ - generated sample, $X_1 \sim p_{\text{data}}$ - real sample, $f_\psi(\cdot)$ - learned displacement field.

$$t \sim U[0, \tfrac{1}{2}], \quad B \sim \text{Bernoulli}(\tfrac{1}{2}), \quad \epsilon \sim \mathcal{N}(0, I).$$

$$X_t = I_t(X_0, X_1) + \sigma_t \epsilon, \qquad \widetilde{X}_t = \begin{cases} X_t, & B = 0, \\ X_{1-t}, & B = 1. \end{cases}$$

I_t - symmetric interpolant, σ_t - noise. The field observes $(t, \widetilde{X}_t)$ and the target is always $X_1 - X_0$.

Minimax loss function

$$\min_\theta \max_\psi \mathcal{L}_{\text{MGM}}(\theta, \psi) = \int_0^{1/2} \mathbb{E}\Big[2\langle f_\psi(t, \widetilde{X}_t), X_1 - X_0\rangle - \|f_\psi(t, \widetilde{X}_t)\|_2^2\Big]\, dt.$$

Our adaptation

$$X_1 - \ 128 \times 128, \quad X_0 - \ 32 \times 32, \qquad X_0 = DX_1, \qquad D = \mathrm{AvgPool}_{4\times 4}$$

$$X_1 = G_\theta(X_0, Z)$$

The observation $\widetilde{X}_t$: a randomly flipped mixture of two HR endpoints

$$\widetilde{X}_t = \begin{cases} (1-t)X_0 + tX_1, & B = 0, \\ tX_0 + (1-t)X_1, & B = 1 \end{cases}$$

- The field observes $(X_0, t, \widetilde{X}_t)$ and predicts $Y - X_0$. The flip B stays hidden.
- The target is conditional matching: $p_\theta(Y \mid C) = p_{\mathrm{data}}(Y \mid C)$.

Generator

Projection follows *GAN Prior Based Null-Space Learning for Consistent Super-Resolution* (AAAI 2023).

$$G(c, z) = \Pi_c(U(c) + h(F_\theta([U(c), z]))), \qquad \Pi_c(y) = y + R(c - Dy).$$

U - bicubic upsampling, R - nearest neighbor repliction, $DR = I$, $D\Pi_c(y) = c$.

Component	Our v1	Our v2
Generator residual h	tanh	identity
Field time input	four spatial time maps	sinusoidal embedding + MLP
Generator / field parameters	16.02/16.03 M	16.02/17.29 M

U-Net widths 120/240/480

Separate training settings: $\sigma_t^{f/G} = \sqrt{\lambda_{f/G} t(1-t)}$.

Data

Training

- DIV2K. 800 training photographs.
- Four newly sampled 128×128 crops per photograph per epoch.
- 3,200 patches per epoch, no augmentations.
- LR. 32×32 average pooling after the HR crop.

Evaluation

- 50 validation and 50 disjoint test photographs.
- Test: five fixed crops per photo, 250 pairs.
- Identical test pairs for every compared method.
- Average crops within photographs, then average 50 photographs.

Baselines

Method	Role and implementation	Inference params
MSE	Same U-Net and projection	16.02 M
MSE + perceptual	Same U-Net + external VGG feature loss	16.02 M
ESRGAN	Standard RRDBNet geometry + our training recipe	16.70 M
I^2SB-small	Our time-conditioned U-Net + bridge sampling	17.29 M
SinSR	Author checkpoint (trained on ImageNet)	173.92 M

Metrics for evaluation:

- **PSNR/SSIM**
- **LPIPS**
- **Latency**

Test quality

Baselines and author reference

Method	PSNR ↑	SSIM ↑	LPIPS ↓	ms
MSE	**31.476**	**0.8071**	0.2244	9.56
MSE + perc.	30.993	0.7967	0.1525	9.58
ESRGAN	27.536	0.7212	0.1456	37.77
I^2SB 1 NFE	30.841	0.7938	0.2514	11.36
I^2SB 4 NFE	28.769	0.7490	0.2437	37.96
I^2SB 16 NFE	27.925	0.7215	0.2112	146.95
I^2SB 64 NFE	27.394	0.6995	0.1928	585.05
SinSR	26.778	0.7021	0.1685	140.21

All MGM settings

Method	PSNR ↑	SSIM ↑	LPIPS ↓	ms
v1 no noise	28.167	0.7249	**0.1387**	9.08
v1 matched .01	27.658	0.7075	0.1464	9.17
v1 critic .01	26.855	0.6594	0.1891	9.18
v1 matched .05	27.546	0.6985	0.1520	9.10
v1 midpoint critic .01	27.676	0.7037	0.1406	9.11
v2 no noise	27.948	0.7161	0.1391	9.13
v2 matched .01	27.650	0.7019	0.1402	9.13
v2 critic .01	27.703	0.7058	0.1423	**8.96**
v2 matched .05	27.525	0.6984	0.1482	9.03
v2 midpoint critic .01	27.626	0.7018	0.1423	9.29

All 18 completed test configurations. I^2SB uses our compact backbone.

Test: 50 photos, five fixed crops each. Local models: mean over three seeds. SinSR: one author checkpoint. All five noise regimes for both MGM architectures. Underlined: MGM. Bold: best mean in each column across both tables.

Test quality and projections

Baselines and author reference

Method	PSNR ↑	SSIM ↑	LPIPS ↓
MSE	**31.476**	**0.8071**	0.2244
MSE + perc.	30.993	0.7967	0.1525
ESRGAN	28.943	0.7371	**0.1372**
I^2SB 1 NFE	31.065	0.7967	0.2417
I^2SB 4 NFE	30.083	0.7749	0.1771
I^2SB 16 NFE	29.455	0.7525	0.1507
I^2SB 64 NFE	28.904	0.7319	0.1428
SinSR	28.340	0.7294	0.1460

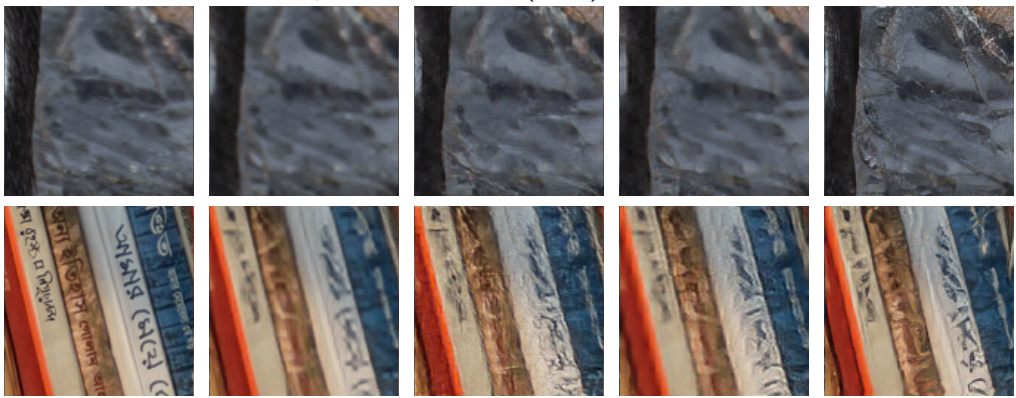
All MGM settings

Method	PSNR ↑	SSIM ↑	LPIPS ↓
v1 no noise	28.167	0.7249	0.1387
v1 matched .01	27.658	0.7075	0.1464
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v2 midpoint critic .01	27.626	0.7018	0.1423

The same final Π_c for every method.

Test: 50 photos, five fixed crops each. Local models: mean over three seeds. SinSR: one author checkpoint. All five noise regimes for both MGM architectures. Underlined: MGM. Bold: best mean in each column across both tables.

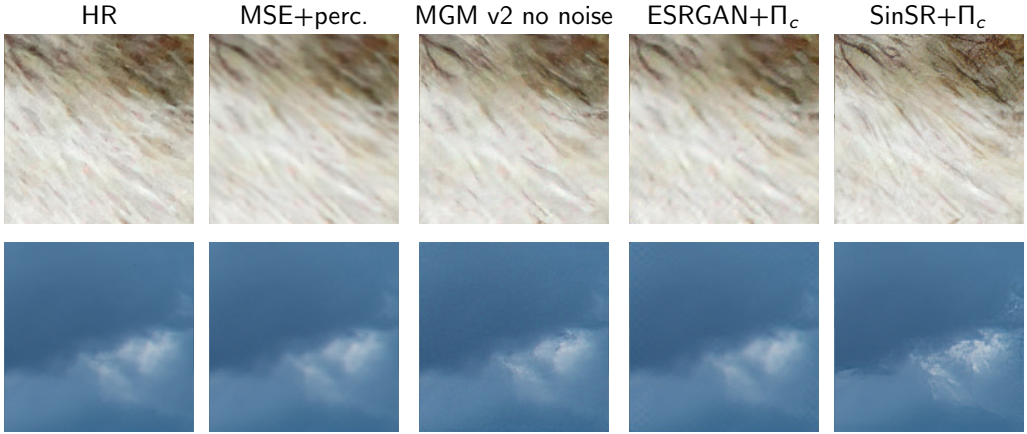
Generated images in test 1/4



Compare texture and lettering against HR. Mean LPIPS does not describe every photograph.

Validation illustrations (not test scores): photographs 0801 and 0867, crop 0; MGM is the original critic-noise control. Local models: seed 0; external SinSR: author checkpoint. Original exported pixels; the second example was selected for visible objects and lettering.

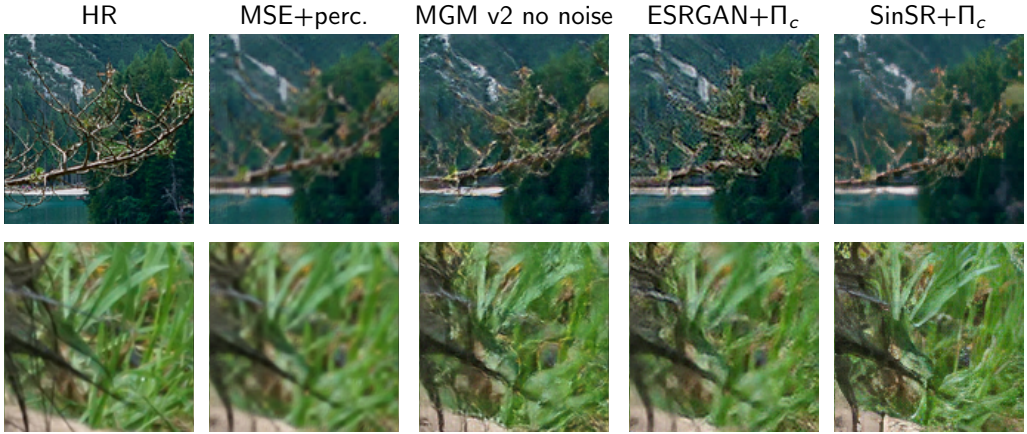
Generated images in test 2/4



Compare texture and structure with HR. Inspect smooth regions for added grain.

Test photos 0805, 0806, crop0, local seed0/draw1. First eight test photos in fixed order. MGM: v2/no-noise. SinSR: author checkpoint. Original exported pixels.

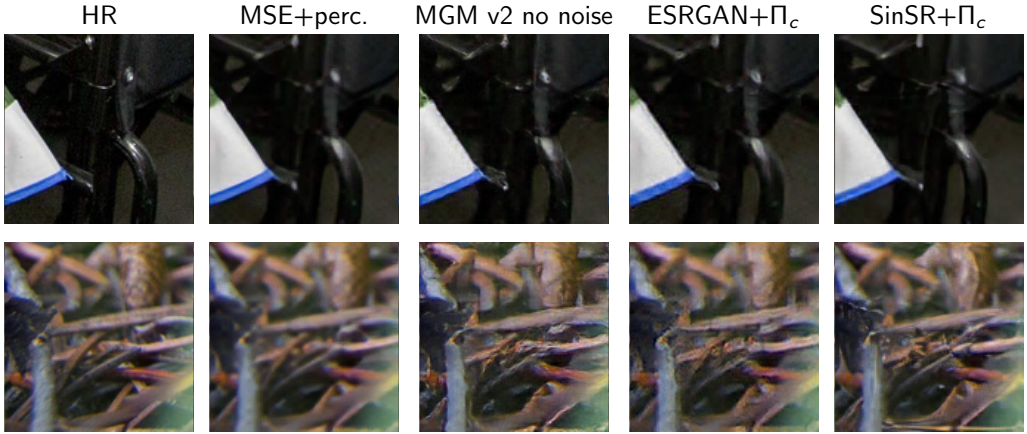
Generated images in test 3/4



Compare texture and structure with HR. Inspect smooth regions for added grain.

Test photos 0807, 0809, crop0, local seed0/draw1. First eight test photos in fixed order. MGM: v2/no-noise. SinSR: author checkpoint. Original exported pixels.

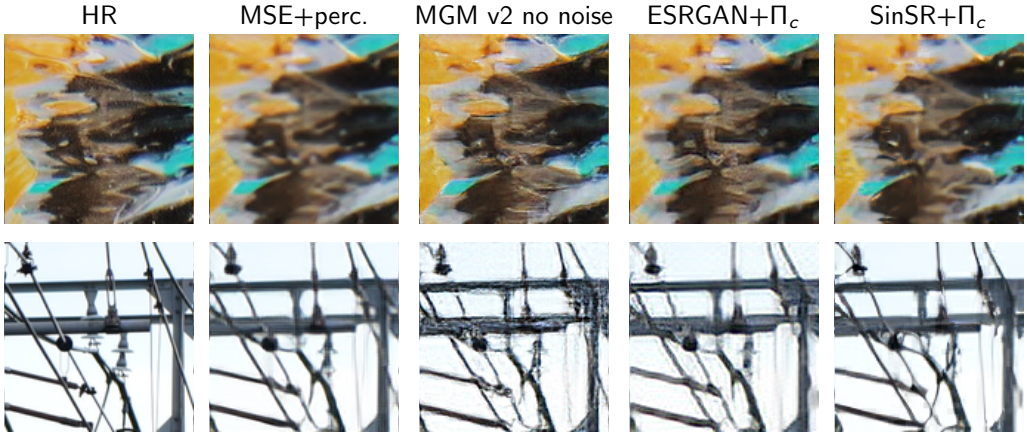
Generated images in test 4/4



Compare texture and structure with HR. Inspect smooth regions for added grain.

Test photos 0813, 0815, crop0, local seed0/draw1. First eight test photos in fixed order. MGM: v2/no-noise. SinSR: author checkpoint. Original exported pixels.

Test illustrations: 4/4



Compare texture and structure with HR. Inspect smooth regions for added grain.

Test photos 0817, 0820, crop0, local seed0/draw1. First eight test photos in fixed order. MGM: v2/no-noise. SinSR: author checkpoint. Original exported pixels.

Limitations

Current limits

- One known degradation and small crops; no proof of conditional distribution recovery. v1/v2 change several factors.
- Training compute is not matched. MGM v2 takes about 2.5× the MSE+perceptual training time.
- VOSR produces structural artifacts in our adapter run; root cause is unresolved.

Research hypotheses, not completed experiments

1. **Adaptive field updates:** use held-out field diagnostics to allocate critic steps under a fixed GPU budget. Test whether stronger fields improve generator quality.
2. **Separate noise roles:** the completed noise sweep favors no interpolant noise here; next isolate latent amplitude and field strength on fixed inputs.

Next controls: warmup $G(c, \alpha z)$ for $\alpha \in \{0, 0.1, 0.5, 1\}$; fixed-compute field-strength and latent-amplitude ablations.

Conclusions and resources

1. Conditional MGM supports a **16 M-parameter, one-pass** SR prototype.
2. Selected v2/no-noise achieves **test LPIPS .1391 at 9.13 ms**, with a useful trade-off against iterative I^2SB-small.
3. **MSE+perceptual and projected ESRGAN remain strong controls.** A general quality advantage has not been established.

Compute: HSE University supercomputer; V100-SXM2-32GB. Audited training series: 89.68 GPU-hours, including the noise sweep; evaluation and earlier attempts excluded.

AI assistance: Astra for analysis, dataset and paper search, and interpretation of research papers. Sol for writing and editing code.
Subscription: ChatGPT Plus.

github.com/kirillkatsuba/mgm_i2i *Sources and reproducibility details accompany the report.*

References

- D. Shlenskii et al. *Midpoint Generative Models*. arXiv:2605.29920v2, 2026.
- X. Wang et al. *ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks*. arXiv:1809.00219, 2018.
- G.-H. Liu et al. *I^2SB: Image-to-Image Schrödinger Bridge*. arXiv:2302.05872, 2023.
- *SinSR: Diffusion-Based Image Super-Resolution in a Single Step*. CVPR 2024. `https://github.com/wyf0912/SinSR`
- R. Zhang et al. *The Unreasonable Effectiveness of Deep Features as a Perceptual Metric*. CVPR 2018.
- E. Agustsson and R. Timofte. *NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study*. CVPR Workshops 2017.
- Y. Wang et al. *GAN Prior Based Null-Space Learning for Consistent Super-Resolution*. AAAI 2023. doi:10.1609/aaai.v37i3.25372
- VOSR official code and checkpoint configuration. `https://github.com/cswry/VOSR`

Backup: why the field regression loss is not the generator loss

The paper's optimal field is $f_t^\star(x) = \mathbb{E}[X_1 - X_0 \mid \widetilde{X}_t = x]$. For fixed t, completing the square gives

$$\mathbb{E}\Big[2\langle f_t(\widetilde{X}_t), X_1 - X_0\rangle - \|f_t(\widetilde{X}_t)\|_2^2\Big] = \mathbb{E}\|f_t^\star(\widetilde{X}_t)\|_2^2 - \mathbb{E}\|f_t(\widetilde{X}_t) - f_t^\star(\widetilde{X}_t)\|_2^2.$$

Evaluated on the same sample and observation, the loss expressions in Algorithm 1 satisfy

$$\widehat{L}_\psi + \widehat{L}_\theta = \|x_1 - x_0\|_2^2.$$

The right-hand side depends on the generator. Minimizing field regression error over θ therefore changes the generator objective. When ψ is frozen, generator gradients must pass through both $f_\psi(t, \widetilde{x}_t)$ and $x_1 - x_0$. For the matched-noise conditional population objective, apply the original definiteness result at almost every c, with its independence, moment and interpolant assumptions. The critic-only recipe uses different noise on the two updates and is a practical modification.

Backup: training recipes and costs

Method	Warmup epochs	Main epochs	min/seed	Main GiB
MSE	50	80	63.62	9.40
MSE+perceptual	50 (shared)	80	70.71	10.02
ESRGAN	50 (L1)	80	59.71	5.87
I^2SB-small	0	130	64.05	9.53
MGM v1	50 G + 50 field	80	174.69	18.16
MGM v2	50 G + 50 field	80	175.10	18.25

MGM: Adam 10^{-5}, $(\beta_1, \beta_2) = (.8, .999)$, EMA .995; MSE warmup 10^{-4}, (.5, .99), batch 16. Main/field batch 32, gradient clip 10, AMP training. No reconstruction penalty in the main MGM objective.

All local generators see 416,000 patches/seed; MGM field-only adds 160,000. Optimizer updates and GPU budgets differ. Shared baseline warmup is counted once in total compute.

Backup: I^{2}SB-small at all tested NFE

NFE (test)	Native PSNR	Native LPIPS	+P PSNR	+P LPIPS	Native ms
1	**30.841**	0.2514	**31.065**	0.2417	11.36
4	28.769	0.2437	30.083	0.1771	37.96
16	27.925	0.2112	29.455	0.1507	146.95
64	27.394	0.1928	28.904	0.1428	585.05
MGM v2, no noise	27.948	**0.1391**	27.948	**0.1391**	**9.13**

The bridge model uses our 17.29 M-parameter U-Net, 1,000 bridge intervals and $\beta_{\max} = .3$. It is trained for 130 epochs on the common pooling data. The original author ADM checkpoint has not been evaluated. Projection improves fidelity; higher NFE improves LPIPS here while lowering PSNR.

Additional projection time is excluded from the timing column.

Underlined: MGM. Bold: best mean in each column of this comparison.

Backup: interpretation and reproducibility

- Three local training seeds; mean $\pm$ sample SD. SinSR has one fixed author checkpoint and no training-seed SD.
- Test intervals use 20,000 photo-bootstrap resamples after averaging crops and three seeds. They condition on fixed checkpoints and draws.
- All 51 local test evaluations and the single SinSR evaluation share crop identities and the evaluation manifest hash.
- Post-processing order: raw output, optional common projection, then clipping for quality metrics. Raw and clipped consistency errors differ.
- Archives contain logs, configurations, metrics and exported images; checkpoints remain on the cluster. Training Git provenance is unavailable in some records.
- Full project compute, including failed/earlier attempts, is not yet reconstructed. The 89.68 GPU-hour training subtotal is not the full cost.

Backup: MGM noise ablation on test

Architecture	Noise regime	PSNR ↑	SSIM ↑	LPIPS ↓
v1	No interpolant noise	**28.167**	**0.7249**	**0.13872** ± 0.00047
v1	Matched .01	27.658	0.7075	0.14641 ± 0.00645
v1	Critic-only .01	26.855	0.6594	0.18906 ± 0.07553
v1	Matched .05	27.546	0.6985	0.15197 ± 0.00379
v1	Midpoint critic .01	27.676	0.7037	0.14060 ± 0.00463
v2	No interpolant noise	27.948	0.7161	0.13908 ± 0.00238
v2	Matched .01	27.650	0.7019	0.14021 ± 0.00142
v2	Critic-only .01	27.703	0.7058	0.14226 ± 0.00168
v2	Matched .05	27.525	0.6984	0.14819 ± 0.00451
v2	Midpoint critic .01	27.626	0.7018	0.14234 ± 0.00113

Test: 50 photographs, five fixed crops each; means over three seeds, LPIPS ± sample SD. No interpolant noise does not remove latent Z.

Underlined: MGM. Bold: best mean per column; test does not define the selection rule.